



A Discrete Crack Element (DCE) method for the uniform motion of general-shape crack distributions in finite domains

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ABSTRACT

We present a computational methodology for the numerical simulations of spatially-distributed fracture processes in solids subjected to external thermomechanical forces. This technique utilizes discrete Volterra dislocation dipoles, enabling the representation of all elastic properties of complex crack geometries, such as energy, displacement, strain, and stress fields. In two-dimensional fracture analysis, we developed an innovative element called the discrete crack element (DCE) to represent crack distributions with various shapes and orientations. This approach overcomes some of the major limitations of traditional fracture mechanics methods in the Finite Element (FE) framework, first by removing the need for remeshing when cracks propagate and then by increasing the accuracy of the elastic field and stress intensity of cracks near surfaces and other cracks. We demonstrate the numerical precision of this method and show its effectiveness in solving mixed-mode fracture in finite geometries under conditions of uniform motion. We apply the technique to show that the propagation path of a center crack in a compressed disk agrees well with the experiments. Using the minimum elastic energy and the maximum normal stress criteria for crack growth indicated that both are consistent with experiments with no noticeable differences.

1. Introduction

Understanding the onset and growth of cracks is crucial to evaluate the reliability of structures. Linear elastic fracture mechanics has seen extensive use (Rice, 1988), where the deformation at the tip of a crack is divided into three independent modes, and the stress field around the tip is characterized by a stress intensity factor. Furthermore, the potential for crack propagation is evaluated based on the magnitude of the stress intensity factor, which makes it a key parameter in linear fracture mechanics. Several analytical investigations have suggested formulas for stress intensity factors for different types of cracks (Shi et al., 1962; Ishida, 1971; Raju and Newman Jr., 1979; Sanford and Dally, 1979; Newman Jr. and Raju, 1981). However, applying linear elastic fracture mechanics to a multitude of real-world problems is limited by the constraints related to the crack's geometry and the boundary conditions of the material surrounding the crack. Thus, to extend the application of linear elastic fracture mechanics to a broader range of practical problems, a more general representation of fracture must be considered.

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The finite element method (FEM) is a typical computational technique that has been widely applied to fracture problems (Chan et al., 1970; Parks, 1974; Lin and Mar, 1976; Rybicki and Kanninen, 1977). A crack is represented by a discontinuity between elements, and the displacement field in a domain containing the crack is then computed. The strain and stress fields are obtained by differentiating the field and using the stress–strain relationships. In linear fracture mechanics (LEFM) using FEM, the stress intensity factor is calculated using computational methods, such as the displacement extrapolation method (Chan et al., 1970; Zhu and Smith, 1995; Guinea et al., 2000), the J-integral method (Rice, 1968), and the virtual crack closure method (Rybicki and Kanninen, 1977; Krueger, 2004). Some of these stress-intensity factor calculations require restrictions on element partitioning near the tip of the crack. In crack growth simulations using FEM, it is necessary to update the element partitioning of the entire region as well as the crack geometry (Trädegård et al., 1998; Bouchard et al., 2000).

To reduce the effort required for such complicated element partitioning, the extended finite element method (XFEM) (Moës et al., 1999) and the s-version FEM (sFEM) (Fish, 1992) have been proposed. XFEM has been used for many types of cracks, such as cohesive cracks (Zi and Belytschko, 2003), fatigue crack propagation (Bergara et al., 2017; Rege and Lemu, 2017), and cracks in bimaterial interfaces (Bouhala et al., 2013; Dimitri et al., 2017). The method does not require element partitioning along the crack because discontinuities in the displacement of the crack face and around the crack tip are embedded within the element by enriching the shape function. Many enrichment functions have been suggested to reproduce the various types and conditions of cracks (Chahine et al., 2008; Bouhala et al., 2013; Wang et al., 2017). However, sFEM uses the principle of superposition to separate the entire domain from the cracked portions, where the domain is modeled in a global mesh. In most cases, the boundary conditions are given in the global mesh, while the crack and its surrounding area are modeled in a local mesh (Kikuchi et al., 2014). The local mesh can be generated separately from the global mesh and can be generated without considering smooth element connections from the tip of the crack to locations far from the tip of the crack, drastically reducing the effort required for the partitioning of the elements along the cracks. This local mesh can then be superimposed on the global mesh to obtain the deformation and stress state of the cracked object. One of the drawbacks of FE-based methods is that although the stress intensity factor can be calculated using the numerical solution to the crack problem, the displacement and stress fields in the immediate vicinity cannot be accurately calculated because of field singularities at the tip of the crack.

In contrast to FE-based approaches that require mesh generation for the entire domain, computational methods that discretize only cracks have been proposed. The boundary element method (BEM) (Blandford et al., 1981; Brebbia and Dominguez, 1994) is a representative computational method of this kind. Nevertheless, the BEM is only suitable for solving crack problems with symmetry conditions. To avoid this limitation, domain decomposition techniques (Lin et al., 1996) and dual BEM (Portela et al., 1992; Mi and Aliabadi, 1992) have also been proposed. In the domain decomposition technique, the domain containing cracks is divided into several subdomains such that the two overlapping crack surfaces are in different regions. However, the shape and position of the boundary for domain decomposition are highly arbitrary, making the boundary setup cumbersome. Moreover, discretizing the boundaries necessary for domain decomposition increases the degrees of freedom of the problem. In the dual BEM, two different types of equations are applied to each overlapping crack face. One is a displacement equation, and the other is a traction equation. This enables us to solve crack problems without the symmetry condition. However, it is still difficult to calculate the traction equation because it has a hyper-singularity (Gallego and Domínguez, 1996; Richardson et al., 1997).

Some methods similar to BEM have also been proposed using the concept of dislocation distributions (Bilby et al., 1963; Weertman, 1996), such as dislocation density (Cleveringa et al., 2000) and dislocation dipoles (Dai, 2002). Unlike BEM, the dislocation density technique does not discretize the crack face but rather discretizes the crack itself. In other words, the crack is treated as a single line (plane in 3D) and then discretized. The integrated value of the dislocation density along the crack plane is the crack opening displacement. The dislocation density distribution is determined from the condition that the crack must be traction-free if the crack is not internally pressurized. Similarly to the traction equation in the dual BEM, this approach also has a singularity. To calculate the singular dislocation density equation, the Chebyshev type of numerical integration (Theocaris and Ioakimidis, 1977) is commonly used. However, the method that uses dislocation dipoles (sometimes called the discontinuous displacement method) assumes a dislocation dipole for each element that discretizes the crack and determines the magnitude of the Burgers vector of the dislocation dipole using the condition of zero traction at the crack surface. As in the method using dislocation density, the equations have singularities that require Chebyshev-type numerical integration. In addition to the problem of the singularity of the equation, the methods using dislocation density and dislocation dipoles specialize in discretizing cracks and require ingenuity to handle finite bodies and complex boundary conditions.

In contrast to the methods of dislocation density and dislocation dipoles, we have developed a numerical solution for cracks using discrete dislocations (Takahashi and Ghoniem, 2013; Sheng et al., 2019). In our method, discrete Volterra dislocation dipoles (or 3D dislocation loops) are arranged on the crack face and the position of dislocations is updated so that the traction, which is the mechanical condition of the crack face, is zero at each dislocation position. To update the dislocation coordinates, the forces acting on the dislocations are calculated using the Peach-Koehler equation. When the force calculated by the Peach-Koehler equation is zero, the mechanical condition of the crack face is satisfied, and the final balanced position of the dislocation is calculated by moving the dislocation according to the dislocation dynamics (DD) method. This method satisfies the mechanical condition of the crack face by updating the position of the dislocation with a constant Burgers vector, whereas the method using the dislocation density or dislocation dipole is based on the magnitude of the Burgers vector of the dislocation density or dislocation dipole at a discretized point. Furthermore, the method using discrete dislocations can solve the cracking problem easily, rather than the methods with dislocation density and dislocation dipoles, since the DD method needs to calculate the stress values at the position of discrete dislocation locations without singular integrals. However, although the numerical method based on dislocations can handle

only planar cracks, it cannot deal with complex changes in crack geometry, such as bending or branching of the crack as the crack propagates.

In our previous studies, we employed the parametric dislocation dynamics (PDD) framework to model three-dimensional cracks in materials (Takahashi and Ghoniem, 2013; Sheng et al., 2019), with a particular emphasis on penny-shaped geometries. Takahashi and Ghoniem (2013) analyzed the quasistatic fatigue propagation of penny-shaped cracks using Paris law as the growth criterion, while Sheng et al. (2019) extended PDD to embedded and surface cracks, illustrating its application to a surface crack that penetrates a beam under pure bending. However, these formulations restricted the cracks to remain planar and assumed that their propagation speeds were low enough to neglect any elastic field distortion. Furthermore, the applications of the method in Refs. Takahashi and Ghoniem (2013), Sheng et al. (2019) were for single cracks that do not interact with other cracks or branch into new configurations. In the present work, we remove these limitations, where we consider non-planar cracks that can propagate along curved paths, interact with other cracks, and move at high speeds. We present here a computational method for studies of fracture processes in brittle solids subjected to external thermomechanical loads. The method is based on the distributions of discrete Volterra dislocation dipoles that can represent all elastic features of complex-shaped cracks, including energy, displacements, strain, and stress fields. A new type of element, which we term the Discrete Crack Element (DCE), is developed. The DCE enables the representation of crack distributions of general shape in 2-D fracture problems. The approach overcomes the challenges of current fracture mechanics methods within the FEM Framework, such as the need for re-meshing of moving cracks and often lack of accuracy close to the crack tips. In addition, the methods using BEM, dual BEM, dislocation density, and dislocation dipoles require the treatment of singular and hypersingular integrals. In contrast, our current method is based on the distribution of discrete Volterra dislocation dipoles and thus does not need handling singular integrals. The objective of this work is to develop and validate the method for various types of cracks, and then show its practical use in an application to the fracture of compressed disks.

This paper is structured as follows. Section 2 outlines the principles underlying our numerical approach for crack problems, utilizing discrete Volterra dislocation dipoles (or loops in 3D), and discusses the variational principle of crack propagation. Section 3 provides a comprehensive description of our numerical methodology. In Sections 4 and 5, we validate our numerical approach by comparing our results with established solutions for kinked cracks, curved cracks, and uniformly moving cracks. Additionally, we demonstrate the practical application of our method to a disk compression problem (Brazilian disk test) by integrating the finite element (FE) solution of the compressed disk with the DCE developed in Section 5. Finally, Section 6 presents our concluding remarks.

2. Equations of motion

Crack motion with uniform velocity in non-conservative (dissipative) materials is obtained from the Principle of Maximum Entropy Production (PMEP), which applies to thermodynamic systems particularly far from equilibrium (Po and Ghoniem, 2014), and without inertial forces. These systems evolve in such a way as to maximize their entropy production rate under constraints. Let us assume that the system is isothermal, then Duhem's inequality states:

$$-\dot{\psi} + \sigma_{kj} \dot{u}_{j,k} = T \dot{s} \geq 0 \quad (1)$$

where ψ is the Helmholtz free energy density, σ_{jk} is the Cauchy stress tensor, $\dot{u}_{j,k}$ is the distortion rate $= \dot{u}_{ij}^E + \dot{u}_{ij}^C$, T is the absolute temperature, and $\dot{s}$ is the internal entropy production rate, and $\dot{u}_{ij}^E$ is the elastic distortion rate tensor and the crack distortion rate tensor $\dot{u}_{ij}^C$. The explicit form of $\dot{u}_{ij}^C$ is obtained by taking the time derivative of u_{ij}^C and applying the surface transport and divergence theorems (Po and Ghoniem, 2014; Slattery et al., 2007). By doing so, one obtains the following.

$$\dot{u}_{ij}^C = \oint_{LC} \delta(\mathbf{x} - \mathbf{x}') b_i(\mathbf{x}') \epsilon_{jkm} V_k d\mathbf{l}'_m + \int_{SC} \dot{b}_i(\mathbf{x}') \delta(\mathbf{x} - \mathbf{x}') da'_j \quad (2)$$

where $\mathbf{b}(\mathbf{x}')$ is the crack-tip Burgers vector, and $\dot{\mathbf{b}}$ is a “crack opening displacement flux” vector, and $\mathbf{V}$ is the crack velocity. Here, the surface bounded by the crack tip line is denoted SC . Substituting into Eq. (1) gives:

$$-\dot{\psi} + \sigma_{kj} \left(\dot{u}_{jk}^E + \dot{u}_{jk}^C \right) = T \dot{s} \geq 0 \quad (3)$$

If we take ψ to be of the form: $\psi(\mathbf{x}, t) \equiv \psi(\mathbf{u}^E)$, then taking the material time derivative gives: $\dot{\psi} = \frac{\partial \psi}{\partial u_{ij}^E} \dot{u}_{ij}^E$. Using the relationship between stress and ψ , the Clausius-Duhem statement then takes the form:

$$\frac{\sigma_{ij}}{T} \dot{u}_{ij}^C = \dot{s} \geq 0 \quad (4)$$

where the internal entropy production rate is $\dot{s}$. If we now assume that cracks will dissipate the entropy produced in the elastic medium $\dot{s}$ by moving at a speed $\mathbf{V}$ such that an entropy dissipation potential $\dot{\phi}$ is a quadratic function of $\mathbf{V}$, i.e. $\dot{\phi} \sim B_{ij} V_i V_j / T$, which also balances out $\dot{s}$, which gives an equality: $\dot{\phi} \equiv \frac{\sigma_{ij}}{T} \dot{u}_{ij}^C = \dot{s}$. B_{ij} is the “drag coefficient” tensor. Using the method of Lagrange multipliers, a constrained functional Π can be written as:

$$\Pi = \int_{\Omega_0} [\dot{s} + \lambda (\dot{s} - \dot{\phi})] dV \quad (5)$$

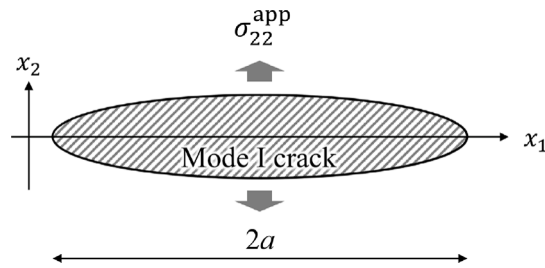


Fig. 1. Schematic of mode-I crack. The x direction is parallel to the crack surface and the x_2 direction is perpendicular to the x direction. A tensile stress σ_{22}^{app} is applied in the x_2 direction.

The functional Π is maximized when $\delta\Pi = 0$ and the value of the Lagrange multiplier λ is determined by $\lambda = n/(1 - n)$ where n is the order of $\dot{s}$ in generalized fluxes, $n = 2$ for quadratic forms. Hence, $\Pi = \int_{\Omega_0} [2\dot{\phi} - \dot{s}] dV$. Substituting the expressions for $\dot{\phi}$ and $\dot{s}$, as well as the value of the Lagrange multiplier $\lambda = -2$, we have:

$$\Pi[\mathbf{V}, \dot{\mathbf{b}}^C] = \int_{\Omega_0} \left\{ \frac{\sigma_{ij}}{T} \int_{SC} \dot{b}_i(\mathbf{x}') \delta(\mathbf{x} - \mathbf{x}') da'_j + \frac{\sigma_{ij}}{T} \oint_{LC} \delta(\mathbf{x} - \mathbf{x}') b_i \epsilon_{jkm} V_k dl'_m - \int_{LC} \frac{1}{2T} \delta(\mathbf{x} - \mathbf{x}') B_{ij} V_i V_j dl'_m \right\} dV \quad (6)$$

Using the sifting property of the Dirac δ function, the functional Π may be rewritten as:

$$\Pi[\mathbf{V}, \dot{\mathbf{b}}] = \int_{SC} \frac{\sigma_{ij}}{T} \dot{b}_i n_j da + \oint_{LC} \left[\frac{\sigma_{ij}}{T} (b_i \epsilon_{jkm} V_k) - \frac{1}{2T} B_{ij} V_i V_j \right] \hat{\xi}_m dl \quad (7)$$

Noting that $\sigma_{ij} n_j = 0$ and taking the variation of Π with respect to the generalized fluxes ($\mathbf{W}$) and imposing stationarity:

$$\delta\Pi = 0 = \oint_{LC} (\epsilon_{ijk} \sigma_{jm} \dot{b}_m - B_{ij} V_j) \delta V_i \hat{\xi}_k dl \quad (8)$$

The first term is the Peach-Koehler force (PK), describes the configurational change in energy with respect to a change in crack shape or position. The PK force is identical to the J-integral, as we established in our previous work (Sheng et al., 2019). If the crack front is represented as a long Volterra dislocation dipole where the dislocation lines are along the x_3 direction ($\hat{\xi}_k dl = dx_3$) with a fixed Burgers vector (displacement jump) b , the closed line integral can be approximated as two line integrals on each dislocation. Since the variation δw_i is arbitrary and assuming that the drag coefficient tensor is isotropic $B_{ij} = B\delta_{ij}$, we get an equation of motion for each dislocation representing the leading edge of the crack in the simple, yet practically important 2D case (see the general 3D case in Ref. Sheng et al. (2019)).

$$\int B_{ij} V_j dx_3 = B \int \delta_{ij} V_j dx_3 = B \int V_i dx_3 = \int \epsilon_{ijk} \sigma_{jm} \dot{b}_m dx_3 \quad (9)$$

The J-integral of a crack in quasi-equilibrium is thus obtained by calculating the unbalanced PK force acting on the two leading dislocations at the crack front. In this work, we focus on cracks moving with a uniform velocity, for which inertial dynamic effects are ignored. The fracture toughness of a material corresponds to a critical PK resistance force on the leading dislocation. Once the J-integral increases above the critical value, the crack advances in a dissipative medium. When the onset of crack growth is detected, the leading dislocation (which is initially fixed for a static crack) is displaced according to the 2D equation of motion along the i -direction.

$$V_i = \epsilon_{ijk} \sigma_{jm} \dot{b}_m / B \quad (10)$$

If we orient the crack to move along the $i = 1$ (x -direction), use $\epsilon_{1j3} = \delta_{j2}$, and take $k = 3$, we get :

$$V_1 = \epsilon_{1j3} \sigma_{jm} \dot{b}_m / B = \frac{1}{B} \sigma_{2m} \dot{b}_m = M f_1 = M(\sigma_{21} b_1 + \sigma_{22} b_2 + \sigma_{23} b_3) \quad (11)$$

Here, V_1 is the crack-tip velocity in the 1-direction, and f_1 is the 1-component of the PK force. Energy dissipation mechanisms are numerous in brittle materials (point- and extended-defect formation, phonon drag, heat flows, etc.) and will not be discussed further here. All these mechanisms can be lumped into an effective mobility coefficient $M = 1/B$, which may have a physical basis for the leading crack dislocation.

3. Modeling plane crack problems

3.1. Mode-I cracks

Consider a simple problem of a uniform tensile specimen with a center mode I crack, as shown in Fig. 1. Let the width of the crack be $2a$ and the applied uniform tensile stress be σ_{22}^{app} . The traction forces on the crack surfaces must vanish as a boundary requirement.

To model the displacement jump across the surface of a mode-I crack, discrete dislocations are positioned along the crack surface with their Burgers vectors along the x_2 direction as shown in Fig. 2. The stress components induced by a two-dimensional straight edge dislocation are expressed analytically as follows Anderson et al. (2017):

$$\begin{cases} \sigma_{11} = -\frac{\mu b}{2\pi(1-\nu)} \frac{x_1(x_1^2 - x_2^2)}{(x_1^2 + x_2^2)^2} & \sigma_{22} = -\frac{\mu b}{2\pi(1-\nu)} \frac{x_1(x_1^2 + 3x_2^2)}{(x_1^2 + x_2^2)^2} \\ \sigma_{33} = -\frac{\mu \nu b}{\pi(1-\nu)} \frac{x_1}{x_1^2 + x_2^2} & \sigma_{12} = -\frac{\mu b}{2\pi(1-\nu)} \frac{x_2(x_1^2 - x_2^2)}{(x_1^2 + x_2^2)^2} \\ \sigma_{23} = \sigma_{32} = 0 \end{cases} \quad (12)$$

where μ is the elastic shear modulus, ν is the Poisson ratio, and b is the magnitude of the Burgers vector. The direction x_2 is parallel to the direction of the Burgers vector, the direction x_3 is parallel to the direction of the dislocation line, and the direction x_1 is perpendicular to the directions x_2 and x_3 . By arranging these discrete dislocations on the crack surface, the stresses created by the discrete dislocations at the crack surface locations can be used to satisfy the condition of zero traction force. The surface traction is $t_i = \sigma_{ij}n_j$, where n_j is the unit normal vector of the crack surface. In the crack problem shown in Fig. 1, the normal direction of the crack surface is in the x_2 direction; therefore, the tractions acting on the crack surface are: $t_1 = \sigma_{12}n_1$ & $t_2 = \sigma_{22}n_2$. Therefore, both σ_{12} and σ_{22} must be zero to satisfy the condition of zero traction force on the crack surface. Regarding σ_{12} , x_2 in Eq. (12) must always be zero for σ_{12} to be zero on the crack surface. This condition is satisfied by taking the direction of the Burgers vector of the discrete dislocations to be perpendicular to the crack surface, so that $x_2 = 0$ at any location on the crack surface. On the other hand, with respect to σ_{22} , since a uniform tensile stress σ_{22}^{app} is applied in the problem shown in Fig. 1, the stress created by the discrete dislocations on the crack surface must actually be equal to $-\sigma_{22}^{app}$. This condition can be satisfied by arranging discrete dislocations on the crack surface.

Fig. 2(a) shows how the discrete dislocations are distributed so that the Burgers vector is perpendicular to the crack surface. Dislocations located at the tip of the crack represent the size of the crack and are fixed. In this study, these fixed dislocations are called crack-tip (leading edge) dislocations. It should be noted that the size of the Burgers vector indicates the displacement of the crack opening. To obtain a dislocation configuration that satisfies zero surface forces on the crack surface, we use the dislocation dynamics method, where the PK force on a dislocation is first calculated, and then its position is updated according to its velocity. The force acting on the i th dislocation when the Burgers vector is oriented perpendicular to the crack plane (x_1 direction) can be calculated by the following equation.

$$f_1^i = (\sigma_{22}^{app} + \sigma_{22}(x_i)) b_2 \quad (13)$$

where, $\sigma_{22}(x_i)$ is the 22 component of stress at the i th dislocation position x_i created by other dislocations. Iterations of the procedure are performed until the force (or velocity) on each dislocation is negligible. In this condition, from Eq. (13), the stress $\sigma_{22}(x_i)$ created by the dislocation at position x_i is equal to $-\sigma_{22}^{app}$, and the condition is satisfied that the traction force is zero at the location of the dislocation on the crack surface. Under plane strain conditions, the stress intensity factor K can be calculated as follows using the force f_1^i action on the crack-tip dislocation.

$$K = \sqrt{\frac{2\mu g}{1-\nu}} = \sqrt{\frac{2\mu f_1^i}{1-\nu}} \quad (14)$$

where, g is the energy release rate. The energy release rate g is equal to the force f_x^i because the work in the crack growth process is equivalent to that performed by the dislocation of the crack tip.

3.2. Mode-II cracks

Using the same method, we can model the mode-II crack, where a uniform shear stress σ_{12}^{app} is applied. Both σ_{12} and σ_{22} must be zero same as the mode-I crack to satisfy the condition of zero traction force at the crack surface and switch the 1 & 2-indices in Eq. (12). Regarding σ_{12} , since a uniform shear stress σ_{12}^{app} is applied, the stress created by the discrete dislocations on the crack surface must actually be equal to $-\sigma_{12}^{app}$. This condition can be satisfied by arranging discrete dislocations on the crack surface. On the other hand, with respect to σ_{22} , x_1 in Eq. (12) must always be zero for σ_{22} to be zero. This condition is satisfied by taking the direction of the Burgers vector of the discrete dislocations to be parallel to the crack surface. Therefore, the mode-II crack can be modeled as shown in Fig. 2(b). In this model, the force acting on the i th dislocation can be calculated by the following equation.

$$f_1^i = (\sigma_{12}^{app} + \sigma_{12}(x_i)) b_1 \quad (15)$$

where, $\sigma_{12}(x_i)$ is the stress component at the i th dislocation position x_i created by other dislocations.

3.3. Mixed mode cracks

We have previously shown that a mixed mode-II/mode-III crack can be modeled using a single set of crack dislocations (Takahashi and Ghoniem, 2013). However, this is only valid in the special case considered, with Burgers vectors along the x_1 -direction, parallel to the applied shear stress σ_{12} . If an additional shear stress is applied along the x_2 -direction, its contribution will not be considered

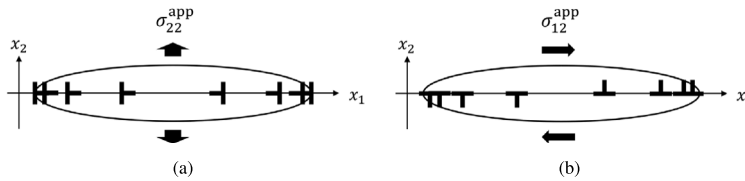


Fig. 2. Schematic of discrete dislocations distributions for (a) mode-I and (b) mode-II cracks.

in computing the Peach-Koehler force. The traction-free crack surface condition with the additional shear stress is: $\sigma \cdot \mathbf{n} = \mathbf{0}$. If only one set of crack dislocations with their Burgers vectors along the 1-direction is used, the magnitude of the glide force given by $f = \mathbf{n} \cdot \sigma \cdot \mathbf{b}$ is completely insensitive to the σ_{23} shear stress component, since:

$$f = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & \sigma_{12} & 0 \\ \sigma_{12} & 0 & \sigma_{23} \\ 0 & \sigma_{23} & 0 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ 0 \\ 0 \end{bmatrix} = b_1 \sigma_{12}$$

For a general 3D loading, three independent sets of dislocations with mutually orthogonal Burgers vectors must be superimposed in order to satisfy the traction-free crack surface condition. For a crack in this configuration, the traction on the crack surface must satisfy $\sigma \cdot \mathbf{n} = \mathbf{0} \rightarrow \sigma_{12} = \sigma_{22} = \sigma_{23} = 0$. For set A, the magnitude of the climb force is given by:

$$f^A = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ b^A \\ 0 \end{bmatrix} = \sigma_{22} b^A$$

Therefore, when the dislocations in set A reach equilibrium, the condition that $\sigma_{22} = 0$ is satisfied. The glide force magnitudes for sets B and C are similarly given by:

$$\begin{aligned} f^B &= \sigma_{12} b_1^B + \sigma_{23} b_3^B \\ f^C &= \sigma_{12} b_1^C + \sigma_{23} b_3^C \end{aligned} \quad (16)$$

Since the Burgers vectors for sets B and C are orthogonal, the only solution which yields equilibrium for both sets is $\sigma_{12} = \sigma_{23} = 0$. Thus, with three sets of dislocations, the traction-free crack surface condition can be satisfied. It should be mentioned that while sets B and C are completely independent of set A, their fields do affect each other. For plane mixed-mode crack (mode-I and II) problems, we use only two sets of edge dislocations simultaneously, since $\sigma_{23} = 0$.

3.4. The discrete crack element for crack propagation

We introduce the Discrete Crack Element (DCE) for problems involving the propagation and interaction of multiple cracks of any shape. The DCE is defined as a small length crack, where several discrete dislocations are distributed inside it. The propagation of a large crack ensemble is modeled by adding DCEs to the crack tip at every simulation step. Fig. 3 shows a schematic of our procedure for crack propagation simulations. Here, we consider a curved crack as the initial crack shown in Fig. 3(a). First, the initial crack is modeled using multiple DCEs (Fig. 3(b)). Since a DCE is defined as a straight crack, small-length DCEs can reproduce a curved crack with a high accuracy as detailed in Section 4.3. Second, the positions of Mode-I and II dislocations in each DCE are calculated using DD (Fig. 3(c)). The algorithm for calculating dislocation positions in a single DCE is shown in Algorithm 1 in Appendix. The applied stress on the crack and the stress generated from other dislocations in the same DCE, as well as the stress generated from all dislocations in other cracks, are used to calculate the force acting on each dislocation. The positions where the force acting on the dislocation (Eqs. (13) and (15)) becomes zero, except for the crack tip dislocation are calculated under the designated Burgers vectors and in the range of the designated number of dislocations. If the force does not become close to zero, the values of the Burgers vector are changed; the Burgers vectors are decreased if the number of dislocations is too large whereas they are increased if it is too small. After the calculation of the dislocation positions, the crack propagation directions are calculated at the crack tip (Fig. 3(d)). In this study, we introduce two criteria to determine the direction of crack propagation: the minimum strain energy density and the maximum normal stress criteria. Finally, new DCEs are added to the crack tip according to the crack propagation direction calculated in the previous step (Fig. 3(d)). By repeating the steps shown in Figs. 3(c), 3(d), and 3(e), our method can reproduce complex crack paths in finite geometry. The pseudo-code for the developed computational method is given in Appendix.

The minimum strain energy density criterion was proposed by Sih in 1974 (Sih, 1974), where the crack propagates in the direction where the strain energy density is minimum. The strain energy density has been calculated using the stress intensity factors at the tip of the crack, the angle, and distance from the tip of the crack in many of the existing crack propagation methods. However, this approach cannot be used for the propagation of cracks in a finite body because it is valid only for cracks in an infinite body under uniform loading. Therefore, we directly calculate the stress field near the crack tip accurately compared to FEM-based methods, and then determine the strain energy density as [gdoutos \(2012\)](#):

$$S = \frac{1}{4\mu} \sigma_{ij} \sigma_{ij} - \frac{\nu}{4\mu(1+\nu)} \sigma_{kk} \sigma_{ll}, \quad (17)$$

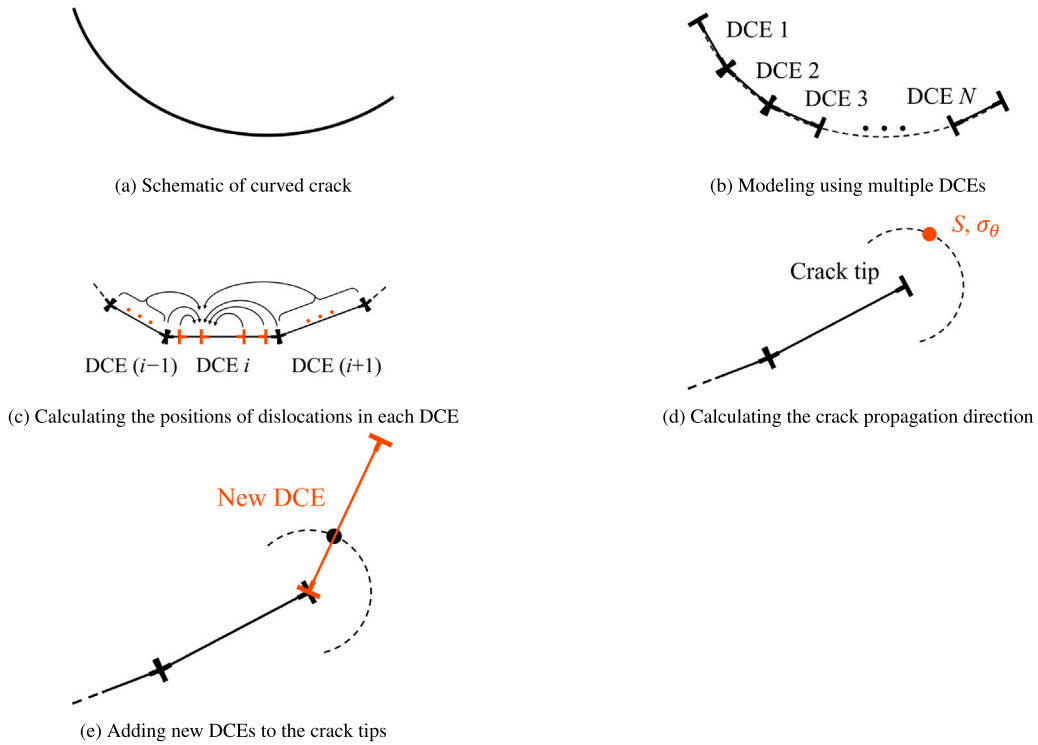


Fig. 3. Schematic of DCE method procedure for crack propagation simulations.

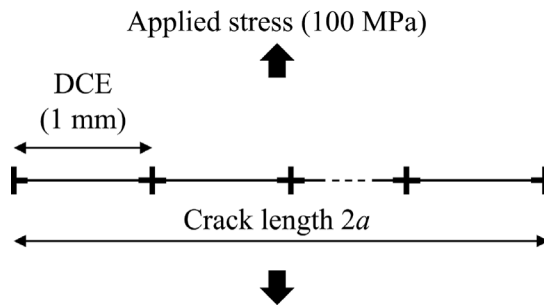


Fig. 4. Schematic of the simulation model of a single straight crack.

where: σ_{ij} is the stress tensor, $\sigma_{kk} = \sigma_{11} + \sigma_{22} + \sigma_{33}$ is the trace of the stress tensor, μ is the shear modulus, and ν is Poisson's ratio.

On the other hand, Erdogan and Sih proposed the maximum normal (hoop) stress criterion (Erdogan and Sih, 1963), where the crack propagates in the direction where the normal stress is highest at the crack tip. In previous studies, the normal stress was calculated using the stress intensity factors, the distance from the tip of the crack, and the crack growth angle. However, because cracks are in a finite body, the normal stress is calculated directly by rotating the stress field near the crack tip:

$$\sigma_\theta = \sigma_{11} \sin^2 \theta + \sigma_{22} \cos^2 \theta - 2\sigma_{12} \sin \theta \cos \theta \quad (18)$$

evaluated at the polar coordinate point (r, θ) with origin at the tip of the crack. In Section 5, we apply these criteria to study crack propagation in a compressed disk. It is noted that the distance between the tip of the crack and the calculation point of the strain energy density and the normal stress does not greatly affect the crack properties if it is set to be approximately 0.05–0.1 of the DCE length.

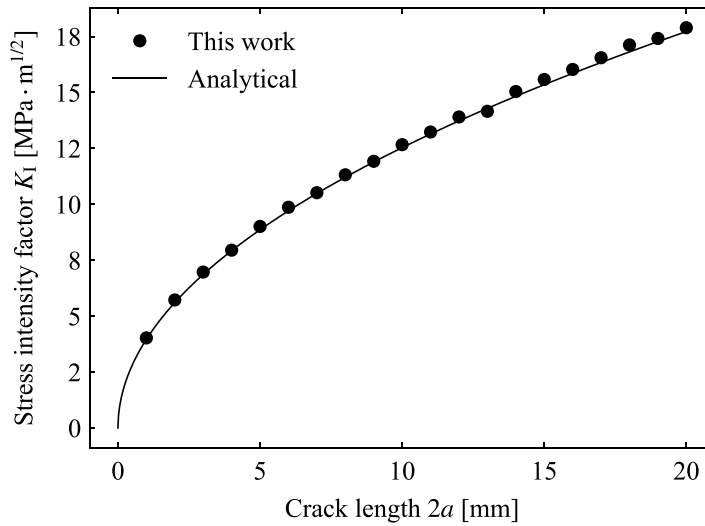


Fig. 5. Relationship between the crack length $2a$ and the stress intensity factor K_I . Analytical results from Ref. Tada et al. (1973).

4. Method validation

4.1. Straight cracks

We illustrate the DCE method by simulation of a single straight crack in an infinite medium, modeled by a number of DCEs, as shown in Fig. 4. A 10 mm crack was modeled using 10 DCEs, each of length 1 mm. The number of dislocations in each DCE was set between 30 and 40. The number, position, and Burgers vector of each dislocation was calculated where the force acting on it becomes small. When the ratio of the force acting on a dislocation to that on the dislocation at the tip of the crack (leading-edge dislocation) falls below 10^{-4} , its position is fixed and considered to be in equilibrium. We calculated the stress intensity factor K_I at the crack tip, applying the mode I stress of 100 MPa. Fig. 5 shows the relationship between crack length $2a$ and stress intensity factor K_I . The analytical relationship is calculated using: $K_I = \sigma\sqrt{\pi a}$, where σ and a indicate the applied stress and the half-crack length, respectively (Tada et al., 1973).

We verify here the accuracy of the numerical method by a direct comparison with the analytical results. Fig. 6 shows the relationship between the crack length $2a$ and the error in the stress intensity factor. The stress intensity factor can be calculated in less than 3% in any crack length. Fig. 7 shows the dislocation distributions and the crack opening displacements U for the crack lengths $2a = 1$ and 6 mm, respectively. The analytical crack displacement, COD (U), is calculated as: $U = \frac{2(1-\nu)\sigma\sqrt{(a^2-x^2)}}{\mu}$, where x is the distance from the center of the crack (Tada et al., 1973). In this calculation, μ and ν were set at 80 GPa and 0.3, respectively. Our result also shows the high accuracy of the COD as compared to the analytical values.

4.2. Kinked cracks

To determine the stress intensity factor of mixed-mode cracks, we used two independent sets of edge dislocations. Here, we focus on how we model a kinked crack as an example, where remote loading results in mixed mode. Fig. 8(a) shows the schematic of a kinked crack, while Fig. 8(b) shows the position of the DCEs. A kinked crack was modeled using two DCEs (DCE 1 and 2), the lengths of which were $a = 2$ mm and b , respectively. The distance c indicates the length projected on the x axis. We calculated the mode I and II stress intensity factors K_I , K_{II} on the crack tip of DCE 2 side for various lengths b and angles θ , which is between the x axis and DCE 2, applying a stress of 100 MPa in the x_2 direction. The number of dislocations in each DCE was set between 20 and 30. Fig. 9(a) shows the dislocation distributions for the angle $\theta = 0^\circ$ and 30° , and the ratio $b/a = 1.0$. Only mode I dislocations are distributed on the crack $\theta = 0^\circ$. On the other hand, both mode-I and -II dislocations are distributed on the $\theta = 30^\circ$ crack because this crack has a mixed mode. Fig. 10 shows the relationship between the angle θ and the stress intensity factor K_I , K_{II} for the ratio $b/a = 0.4$, 1.0, and 2.0. The analytical values are calculated using the method suggested by Kitagawa et al. (1975). The stress intensity factor decreases as the crack is kinked more sharply in both results of mode-I and -II. Our results agree well with the analytical results for any angle θ and ratio b/a .

4.3. Curved cracks

Curved cracks can be modeled as a contiguous assembly of small straight cracks using DCEs. Here, we calculate the stress intensity factor of an arc crack of radius $r = 10$ mm and the angle $\alpha = 80^\circ$ as shown in Fig. 11. The arc crack was modeled using multiple

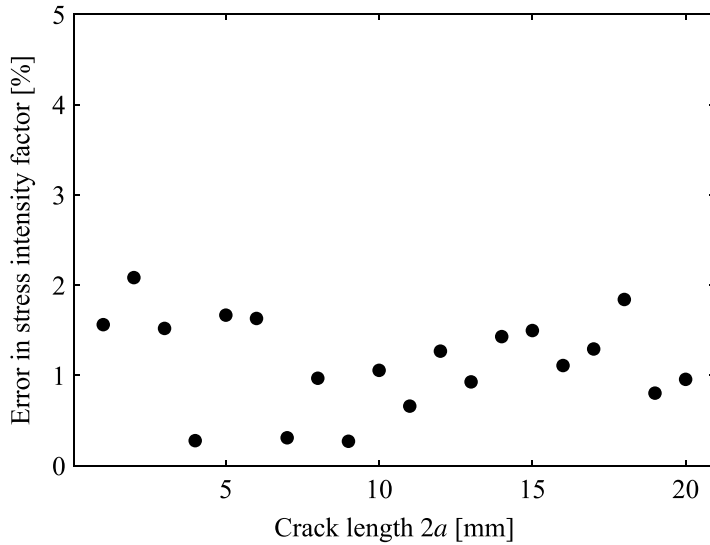


Fig. 6. Relationship between the crack length $2a$ and the error in the stress intensity factor.

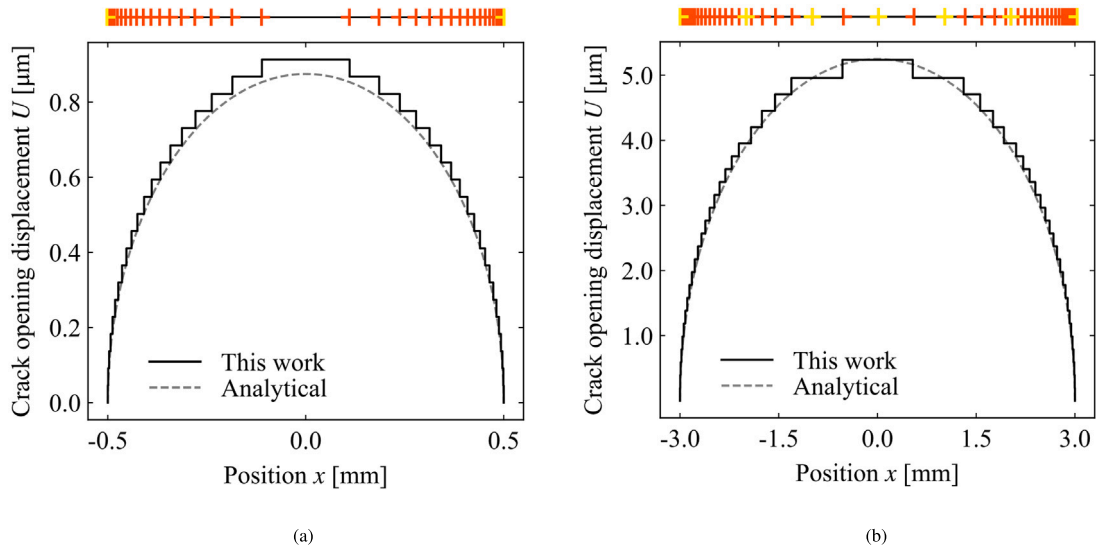


Fig. 7. Dislocation distributions and crack opening displacements U for (a) $2a=1$ mm, and (b) $2a=6$ mm. Yellow and red symbols indicate the crack tip dislocations and the crack dislocations, respectively. Some crack dislocations are invisible because they overlap with crack tip dislocations. The position x is the distance from the center of the crack. Analytical results from Ref. Tada et al. (1973).

DCEs. The applied stress was set to 100 MPa in the x_1 and x_2 directions. The number of dislocations in each DCE was set between 20 and 30. Fig. 12 displays the dependence of the error in the stress intensity factor of mode-I and -II on the number of DCEs and on the ratio of the arc radius to the length of the DCE. The analytical stress intensity factor was calculated using the exact equation, which was originally proposed by Mushkelishvili (1953) and rewritten by Cotterell and Rice (1980). There are error oscillations attributed to the discrete nature of our method. In our approach, the crack geometry is represented by multiple straight crack elements, and the physical behavior of each element is computed based on discrete dislocations. However, the stress intensity factors are seen to converge to the analytical values as the number of DEC's increases, or equivalently, as the ratio of the radius-to-DEC length increases. More than 15 DCEs are required to reproduce the stress intensity factors of mode I and II with a high accuracy with an error of less than 5%, which corresponds to a radius-to-DCE length of around 10. Although the increasing number of discrete dislocations can reduce errors, it may lead to an increase in the computational cost for the dislocations to be converged.

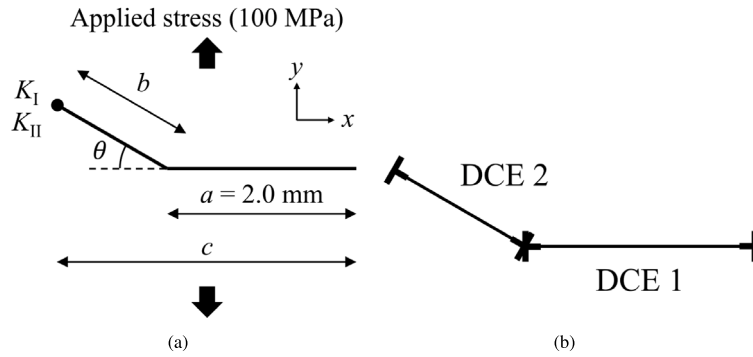


Fig. 8. Schematic of a kinked crack simulation model: (a) Overview of the kinked crack, and (b) DCE distribution.

- └┐ Crack tip dislocation (immobile), mode-I └┐ Crack dislocation, mode-I
- └┐ Crack tip dislocation (immobile), mode-II └┐ Crack dislocation, mode-II

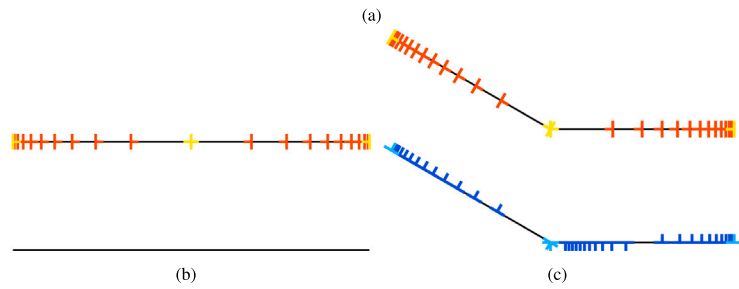


Fig. 9. Dislocation distributions for the angle $\theta =$ (b) 0° and (c) 30° . The distributions of mode-I and II dislocations are shown in the upper and lower rows, respectively. Some crack dislocations are invisible because they overlap with crack tip dislocations.

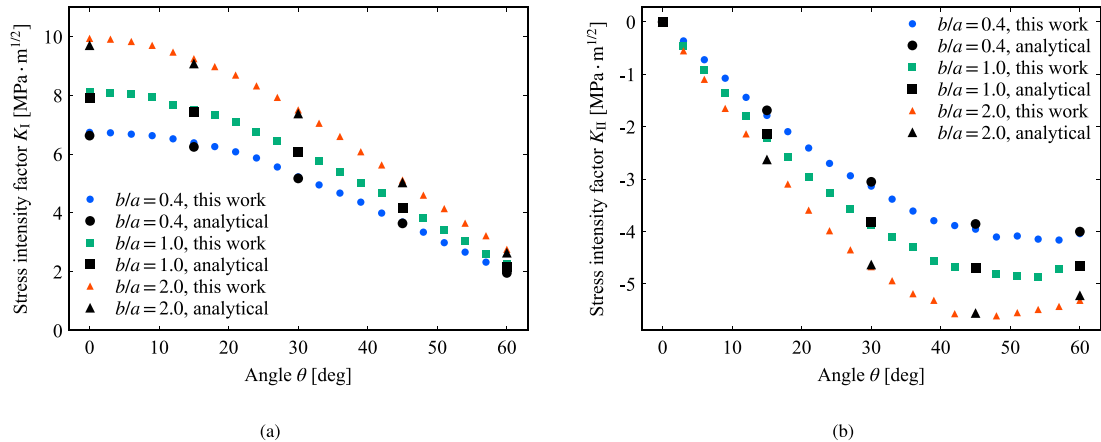


Fig. 10. The relationship between the angle θ and the stress intensity factors (a) K_I and (b) K_{II} for ratio $b/a = 0.4, 1.0$, and 2.0 . Analytical results are by Kitagawa et al. (1975).

4.4. Moving Griffith cracks with uniform velocity

We consider cracks that move at a uniform velocity without acceleration forces. The J-integral and the corresponding PK forces are the same as static cracks, since inertial forces are absent. However, the stress field is greatly influenced by crack motion resulting from coordinate transformations between a moving and a static frame of reference. We validate here our computational method for cracks moving at a uniform velocity, specifically the Griffith crack that moves without acceleration or changing of the crack length

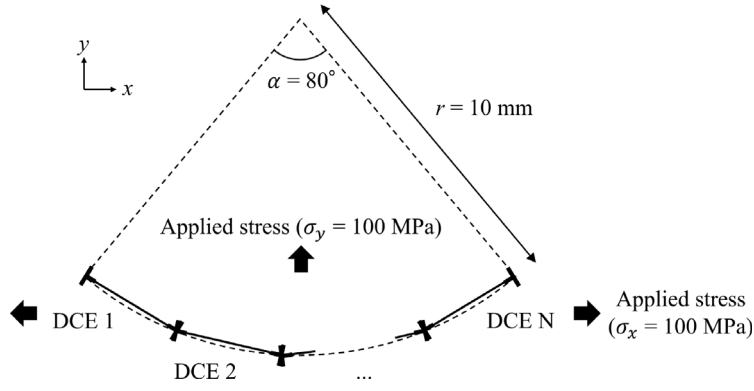


Fig. 11. Schematic of the simulation model of a curved crack.

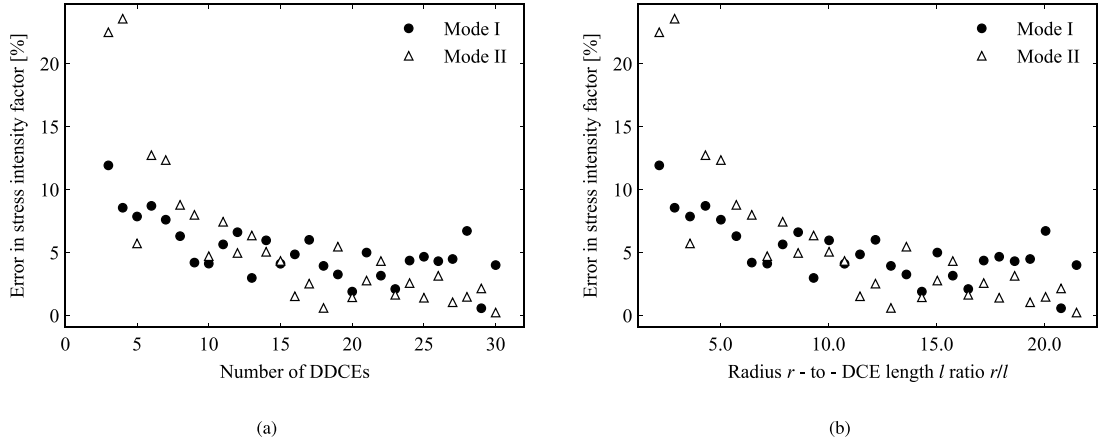


Fig. 12. Dependence of the stress intensity factor error (a) on the number of DCEs and (b) on the ratio of the radius-t-DCE length. The results of both mode-I and II are indicated. Analytical values are calculated using the equation suggested by Cotterell and Rice (1980).

in an elastic medium (Yoffe, 1951). When remote traction is applied on a plate containing a center crack, the crack, which is of length $2a$, is expected to spread in both directions along its length. However, Yoffe (1951) assumed for simplicity that the crack propagates to the right only and remains constant in length. She justified this assumption by the fact that the stress distribution near one end of the crack is not influenced by its distance from the other end (Yoffe, 1951). The analytical stress intensity factors of the Griffith moving crack (Yoffe, 1951) are used for comparison with our numerical method. If the dislocation moves at a uniform velocity V , the stress induced by a dislocation of the two-dimensional straight climbing edge can be expressed as follows (Weertman, 1967):

$$\begin{cases} \sigma_{11} = \frac{2\mu b C_S^2}{\pi V^2 u_L} \left\{ \frac{u_{2S}^2(2u_{2L}^2 - u_{2S}^2)x_1}{(x_1^2 + u_L^2 x_2^2)} - \frac{u_S u_L x_1}{(x_1^2 + u_S^2 x_2^2)} \right\} \\ \sigma_{22} = -\frac{2\mu b C_S^2}{\pi V^2 u_L} \left\{ \frac{u_{2S}^4 x_1}{(x_1^2 + u_L^2 x_2^2)} - \frac{u_S u_L x_1}{(x_1^2 + u_S^2 x_2^2)} \right\} \\ \sigma_{12} = \frac{2\mu b C_S^2}{\pi V^2} \left\{ \frac{u_L u_{2S}^2 x_2}{(x_1^2 + u_L^2 x_2^2)} - \frac{u_S u_{2S}^2 x_1}{(x_1^2 + u_S^2 x_2^2)} \right\} \end{cases} \quad (19)$$

$$\begin{cases} u_S = \sqrt{1 - V^2/C_S^2}, & u_{2S} = \sqrt{1 - V^2/2C_S^2} \\ u_L = \sqrt{1 - V^2/C_L^2}, & u_{2L} = \sqrt{1 - V^2/2C_L^2} \end{cases} \quad (20)$$

$$C_S = \sqrt{\mu/\rho}, \quad C_L = C_S \sqrt{2(1-\nu)/(1-2\nu)} \quad (21)$$

where ρ is the density. C_S and C_L are the transverse and longitudinal sound velocity, respectively. This validation set μ , ν , and ρ to 80 GPa, 0.3, and 7800 kg/m³, respectively. The directions x_1 and x_2 are parallel to the velocity and the Burgers vector directions,

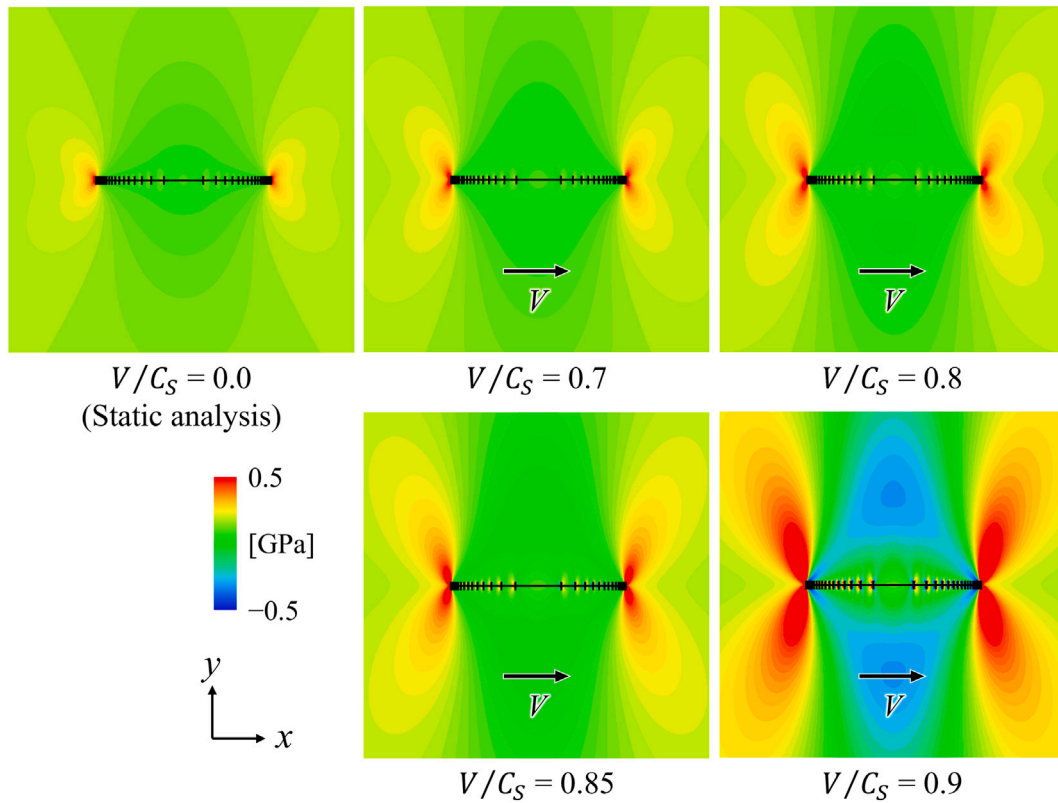


Fig. 13. Stress field σ_y in the y-direction at various crack velocities. The results of the crack velocity $V = 0$ (static analysis), $0.7 C_S$, $0.8 C_S$, $0.85 C_S$, and $0.9 C_S$ are shown. Black symbols on the crack indicate dislocations.

respectively. A mode-I stress of 100 MPa was applied to the crack. Our method can reproduce the Griffith-type moving crack by replacing the stress fields of Eq. (12) with Eqs. (19)–(21).

Here, we modeled a 10 mm Griffith-type crack with various velocities V , between $0.1 C_S$ and $0.9 C_S$. The number of DCE was set to be between 30 and 40, which can reproduce the straight crack properties with a high accuracy shown in Section 4.1. The convergence condition is set to be the same as the previous calculation (i.e. force accuracy of 10^{-4} , relative to the leading-edge dislocation). Fig. 13 shows the stress field σ_y in the y-direction. The stress field became symmetry regardless of the moving direction because the stress field generated by a moving dislocation is not affected by the moving direction indicated in Eq. (19). The stress near the crack tip increases as the crack velocity increases. Fig. 14 shows the relationship between the angle θ of the crack and the normal stress σ_θ . The stress was calculated near the crack tip: the distance from the crack tip was $2a/20 = 0.5$ mm. The stresses calculated by our method are very close to those calculated analytically (Yoffe, 1951), given that the analytical solutions were derived for the limit of a power series in a smallness parameter (k) defining the distance to the crack tip ($k \rightarrow 0$). The stress for $\theta = 60^\circ$ becomes greater as the velocity of the crack increases. Therefore, the crack with high velocity will propagate in the direction of $\theta = 60^\circ$, whereas the crack with a low velocity will propagate in the direction of $\theta = 0^\circ$ (crack direction). Fig. 15 shows the crack opening displacement for various crack velocities. The crack opening displacement increases rapidly as the crack velocity increases. The results using our method are thus consistent with those calculated analytically (Yoffe, 1951). Finally, Fig. 16 shows the relationship between the crack velocity V/C_S and the stress intensity factor K_I . The stress intensity factor increases as the crack velocity increases because the area of a high-stress field increases, as shown in Fig. 13.

5. Fracture of compressed disks

The disk compression test is a compact method used in geomechanics and materials science to assess the compressive behavior and fracture characteristics of disk-shaped specimens. It evolved as a more convenient alternative to traditional compression tests requiring larger, often more difficult to prepare samples. By applying a load along the diameter of the disk, one can get insight into the stress distributions and the growth of cracks in brittle materials under relatively simple boundary conditions. Recent studies have expanded its application to advanced ceramics (Sgambitterra et al., 2018), composites (Budzik et al., 2022), and glassy materials (Chen et al., 2014), often combined with numerical simulations for a deeper understanding. This synergy of experiment and modeling continues to refine how disk compression data inform the design and evaluation of materials under complex loading.

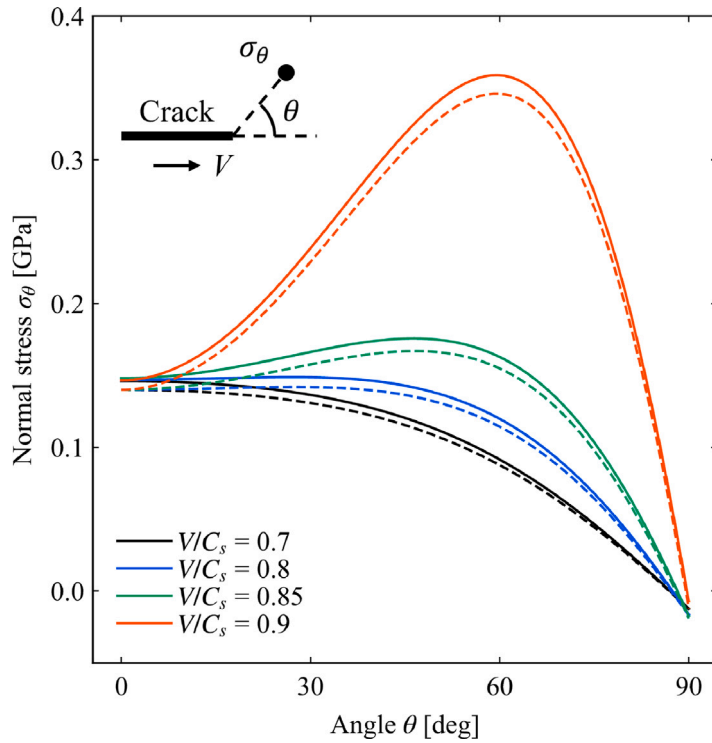


Fig. 14. Relationship between the angle θ from the crack and the normal stress σ_θ for a various crack velocities. The results of the crack velocity $0.7 C_s$, $0.8 C_s$, $0.85 C_s$, and $0.9 C_s$ are shown. The dotted lines indicate the analytical relationships corresponding to Figure (1) of Ref. [Yoffe \(1951\)](#).

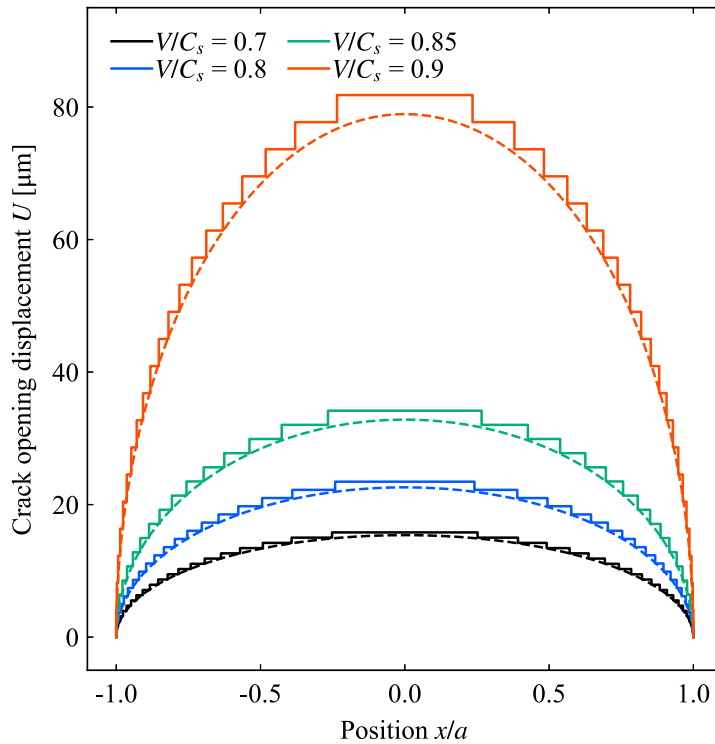


Fig. 15. Crack opening displacements U for various crack velocities. The results of the crack velocity $0.7 C_s$, $0.8 C_s$, $0.85 C_s$, and $0.9 C_s$ are shown. The dotted lines indicate the analytical relationships ([Yoffe, 1951](#)).

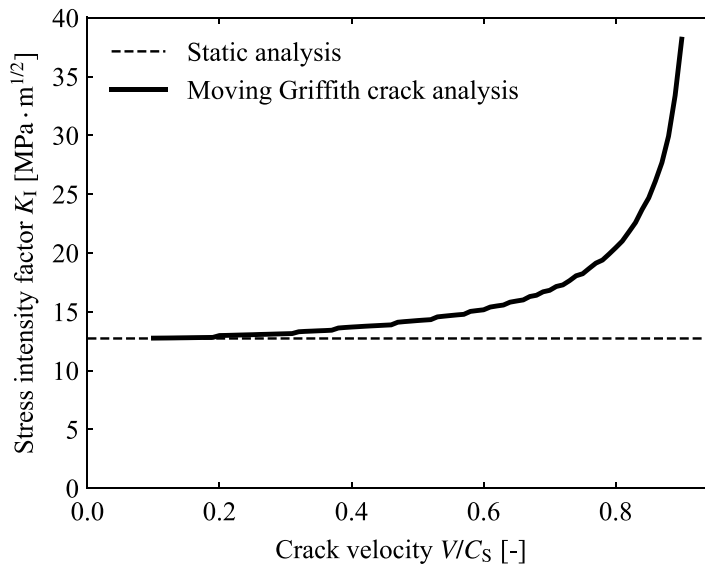


Fig. 16. Relationship between the crack velocity V/C_S and the stress intensity factor K_I . The result of the static analysis was calculated using the method in Section 4.1.

In our previous work, we showed that 3D surface cracks tend to fully penetrate thin disk samples and behave like 2D cracks (Sheng et al., 2019), and it is therefore reasonable to study the growth of 2D cracks in compressed disks. We show here the computational features of our method through an application to the fracture of compressed disks. Shetty et al. (1987) studied the mixed mode fracture of soda-lime glass using a diametral compression test (so-called Brazilian test). The path of a center crack was recorded as it grew from an initial size in the center of the disk to the final fracture in the disk. This experiment is a good test bed for investigating the key mechanism or criterion that determines the direction of crack propagation. Fig. 17 shows the schematic of our simulation model. A soda-lime glass disk with a radius of 25 mm was compressed in the direction of x_1 . The elastic shear modulus and the Poisson ratio of the disk were taken as $\mu = 29.8$ GPa and $\nu = 0.215$, respectively (Duckworth et al., 1983). An initial crack was introduced in the center of the disk with a length of 12.7 mm and an angle from the x_2 axis in the range: $\alpha = 60^\circ - 90^\circ$. The direction of crack growth was determined using two criteria: (1) the minimum strain energy density criterion and (2) the maximum normal stress criterion. In our simulations, a new DCE of length 1 mm was added to the crack tip to satisfy either of the growth criteria with the corresponding orientation. The number of dislocations in the initial DCE and in each added DCE was set between 30 and 40 and between 10 and 20, respectively. The stress field σ_{app} induced by external compression was calculated using the finite element method (FEM). The mesh for FEM is shown in Fig. 17(b). We employed our in-house FE code to calculate linear elastic problems to solve correction problems as described below. The plane strain condition is assumed and 8-node quadratic isoparametric elements are used. The constant stress of 100 MPa was applied to the both side of the disk. The x -direction constraints were added to the bottom and the top of the disk whereas the y -direction constrain was added to the top. We should noted that we just focused on the reproducibility of the crack path in this analysis. Therefore, we did not employ any crack propagation criteria.

The stress field σ_{surf} induced by the interaction between dislocations and the disk surface must be calculated to ensure that the tractions on the disk surface are zero. We calculated the stress field σ_{surf} as shown in Fig. 18. First, surface traction t_∞ in the position of the surface in the infinite body was calculated using Eq. (12). Subsequently, the stress field σ_{surf} was calculated using FEM by adding the surface traction $-t_\infty$ as an external force. In the calculation process to determine the position of each dislocation in each DCE, the stress fields σ_{app} and σ_{surf} were superimposed on the stress field induced by the other dislocations. Details of this superposition procedure can be found in Van der Giessen and Needleman (1995), Sheng et al. (2019).

Fig. 19 shows the stress field $\sigma_{app} + \sigma_{surf}$ induced by applied stress and the interaction between dislocations and the surface, that σ_{dis} induced by dislocations, and the superimposed stress $\sigma_{app} + \sigma_{surf} + \sigma_{dis}$ in the y direction. A tensile stress field is located around the crack by compressing the disk in the x direction (Fig. 19(a)). On the other hand, compressive stress is induced around the crack by dislocations (Fig. 19(b)). By superimposing two stress fields, the stress on the crack becomes almost zero and a tensile stress field is located near the crack tip (Fig. 19(c)). Fig. 20(a) shows the crack growth path calculated by the minimum strain energy density criterion, while 20(b) displays the growth path according to the maximum normal stress criterion. The angle between the initial crack and the crack path was $\alpha = 65^\circ$. The direction of crack growth changes immediately after applying compressive stress. The crack propagates in the x_1 direction, where a mode I stress is induced due to tensile stress in the x_2 direction. In particular, our results of the minimum strain energy density criterion and the maximum normal (hoop) stress criterion show a crack growth path that is highly coincident with that of Shetty's experiment (Shetty et al., 1987). However, the results using existing methods show zigzag paths, which are not experimentally observed. Although the existing methods can almost reproduce the experimental path, the zigzag will lead to generate large errors when either timestep or crack growth rate is set to be large.

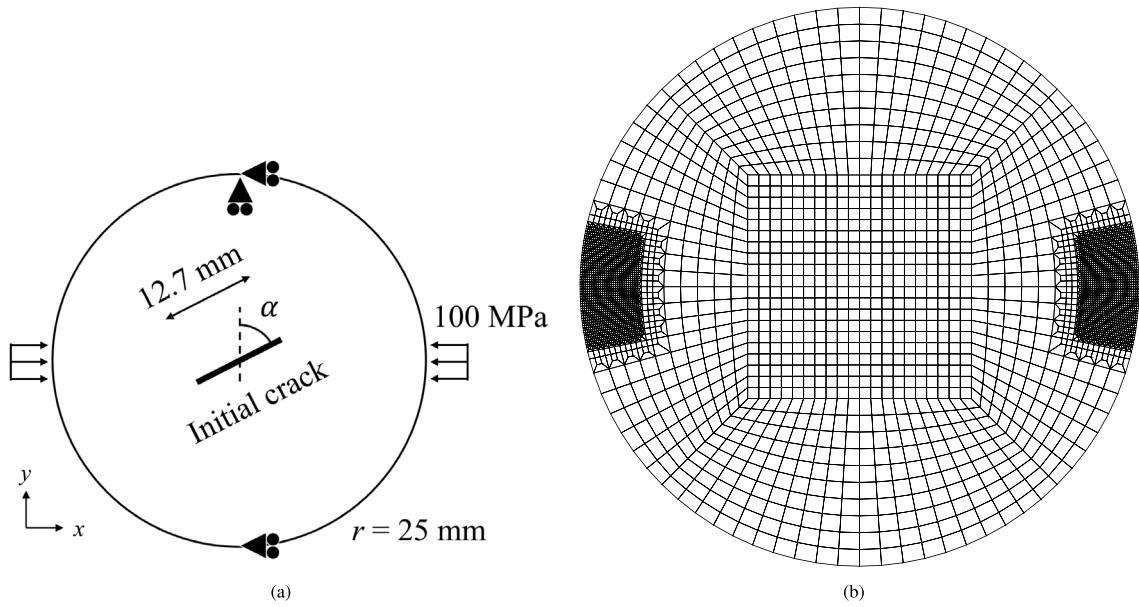


Fig. 17. Schematic of the simulation model of a compressed disk. The compressive boundary stress is in the x_1 direction. An initial crack was introduced in the center of the disk: (a) Overview of the kinked crack, and (b) FE mesh.

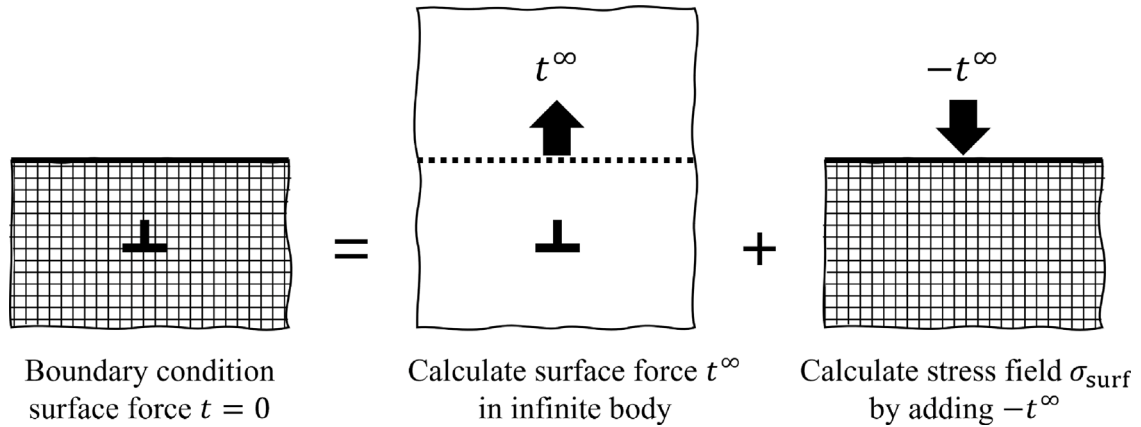


Fig. 18. Calculation method of the stress field σ_{surf} induced by the interaction between dislocations and the disk surface (Van der Giessen and Needleman, 1995; Sheng et al., 2019).

Fig. 21 shows the relationship between the angle of crack growth α and the angle of initial growth of the crack θ . In both results, the relationships are discontinuous around the crack angle $\alpha = 65^\circ$. This is a result of the lack of mode I dislocations for the crack angle $\alpha < 65^\circ$ because the tensile stress in the crack opening direction is small. Using the minimum strain energy density criterion 21(a), the results of our method for the angle θ of the initial crack growth direction (fracture angle) are closer to the experimental values than the analytical methods of Sih (1974), and Erdogan and Sih (Erdogan et al., 1973). In the result of the maximum normal stress criterion (21(b)), our method accurately reproduces the angle θ of the crack growth direction in any initial crack angle α , while the existing methods (Sih, 1974; Erdogan et al., 1973) estimate it to be larger, especially when the initial crack angle α is between 70° and 85° .

6. Discussion and conclusions

A novel computational method has been developed for the analysis of crack propagation in brittle solids. In this approach, cracks are represented by assemblies of discrete Volterra dislocation dipoles, which capture all elastic characteristics of complex crack shapes, including energy, displacement, strain, and stress fields. Central to this work is the introduction of a new element called the Discrete Crack Element (DCE), which makes it possible to represent arbitrarily shaped cracks and their propagation.

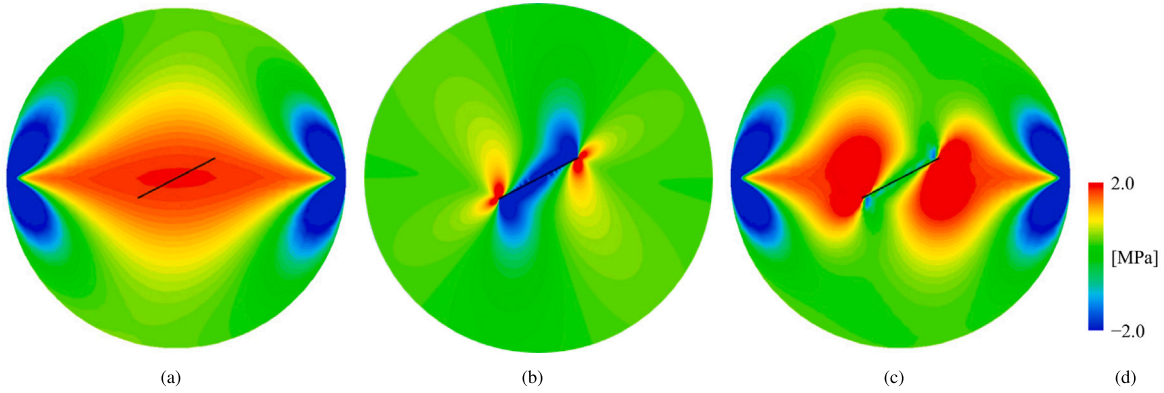


Fig. 19. Stress field component σ_y . (a) stress field $\sigma_{app} + \sigma_{surf}$ induced by applied stress and the interaction between dislocations and the surface. (b) stress field σ_{dis} induced by dislocations. (c) superimposed stress field $\sigma_{app} + \sigma_{surf} + \sigma_{dis}$.

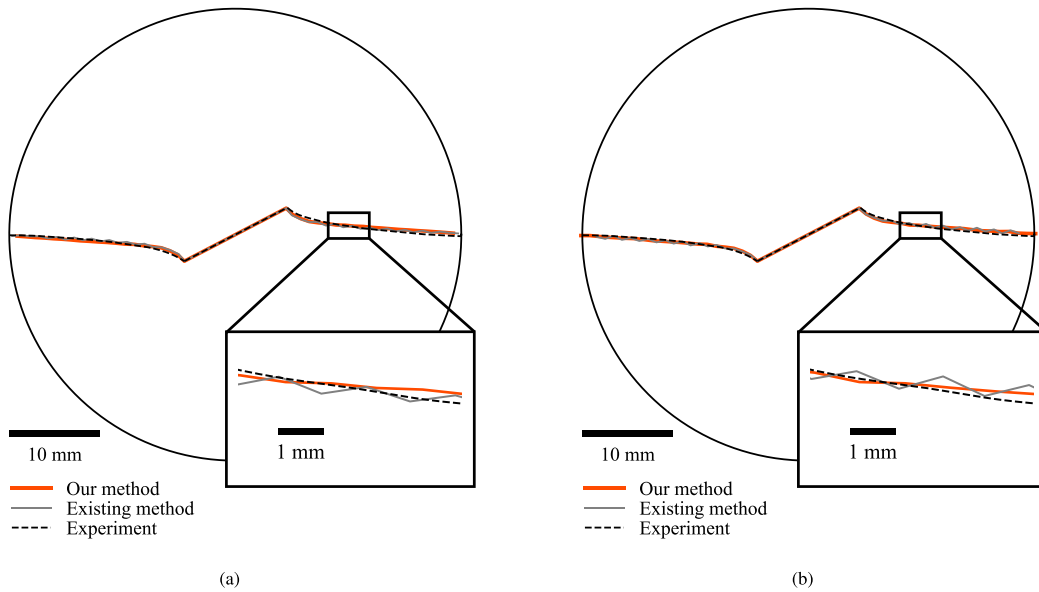


Fig. 20. Crack growth paths calculated by (a) the minimum strain density criterion and (b) the maximum normal (hoop) stress criterion. The experimental results and those calculated by existing methods (Sih, 1974; Erdogan and Sih, 1963).

By arranging two types of edge dislocations, the model can simulate the growth of mixed-mode (mode I and II) cracks within a single framework. It also enables the calculation of stress intensity factors and crack opening displacements, providing comprehensive information on crack behavior. To determine dislocation configurations that satisfy the boundary conditions on the crack surface, we utilize the well-established method of dislocation dynamics. The principles of superposition are used to ensure that the boundary conditions are satisfied in finite geometry, where the FE method is used to calculate a regular elastic field without the singularities associated with Volterra dislocations, and the solution is superimposed on the infinite medium analytical solutions for discrete dislocations. Thus, the finite body does not have to be remeshed every time the crack advances, since the FE solution is regular and is evaluated on the same initial mesh of the domain.

Crack propagation can be modeled by incrementally adding small Discrete Crack Elements (DCEs) at the crack tips. The growth direction is determined using either the minimum strain energy density criterion or the maximum normal stress criterion. Because the stress field near the crack tip is accurately captured, both the strain energy density and normal stress can be reliably calculated, enabling precise predictions of the crack's trajectory and progression. By strategically combining DCEs, the method can accurately reproduce complex crack geometries, such as curved or kinked cracks. Furthermore, quasistatic cracks with constant velocity can be simulated by continuously updating the stress field generated by dislocations as the system evolves. The precision of this

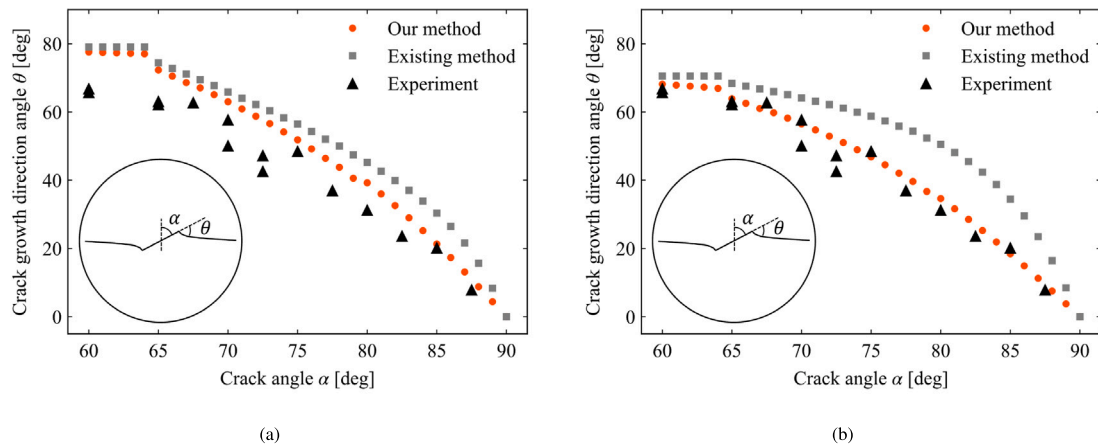


Fig. 21. Relationship between the crack angle α and the initial crack growth direction angle θ calculated by (a) the minimum strain density criterion and (b) the maximum normal (hoop) stress criterion. The results using existing method (Sih, 1974; Erdogan and Sih, 1963) are also shown.

approach has been validated through comparisons with analytical solutions for mixed-mode cracks, kinked cracks, curved cracks, and quasistatic cracks with constant velocity.

One of the main assumptions of the present work is that cracks propagate at constant speed, and thus we neglect crack acceleration. Practically, this means that the inertia forces are negligible and that the driving force (that is, the rate of energy release or the stress intensity factor) remains essentially fixed as it grows. In practice, this is achieved in the overdamped regime (quasistatic), where the inertial term in the 'equation of motion' for the crack tip is tiny compared to the viscous/drag damping. Then, as long as $m \frac{dv}{dt} \ll Bv \implies \tau_{\text{inertial}} = \frac{m}{B} \ll (\text{observation time scale})$, the steady solution $v \approx F_{\text{drive}}/B$ holds almost immediately. The energy release rate G (or stress intensity factor K) seen by the tip must remain essentially constant as the crack progresses. This happens, for example, in a very large specimen under fixed load, where the change in crack length causes only a negligible change in K . Furthermore, the crack velocity v must remain below the Rayleigh wave speed. Thus, dynamic (elastic wave) effects and transient stress field distortions are minimal, so there is no ringing or inertia-driven acceleration. The applied load must not cause sudden jumps (e.g. impact). In cyclic loading with a constant load amplitude ΔK , the Paris law behavior often produces a nearly constant growth per cycle, which, when translated into a per-unit time velocity, is effectively constant over many cycles. Another assumption is that our approach is fundamentally based on the superposition principle to handle the finite geometry of cracked bodies. In other words, it inherently assumes linearity in both material behavior and geometry. Consequently, material and geometrical nonlinearities – such as those arising in ductile fracture – cannot be captured within our framework. Ductile fracture involves large plastic deformation, especially near the crack tip, which necessitates the consideration of both material and geometric nonlinearity in numerical simulations. Therefore, our method is not suitable for modeling ductile fracture. Moreover, phenomena such as plasticity-induced crack closure are directly related to plastic deformation around the crack tip for the same reason. However, we believe that our method can describe crack surface sliding by using glide dislocations without the need for additional complex formulations although surface contact and sliding are often modeled using penalty methods or similar techniques in FEM, which significantly increase the complexity of the simulation.

This approach was used to analyze the propagation of cracks in a diametrically compressed disk, illustrating its practical value. Using straightforward crack growth criteria, the simulated crack trajectories corresponded closely to the experimental observations in Shetty et al. (1987), demonstrating the effectiveness of the method for finite domains. Moreover, the crack paths predicted under both the minimum elastic energy and maximum normal stress criteria aligned with the experimental data, showing no noticeable differences between these two mechanisms.

CRedit authorship contribution statement

A. Hirano: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **A. Takahashi:** Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **M. Alabdullah:** Writing – original draft, Investigation. **N.M. Ghoniem:** Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: N. M. Ghoniem reports financial support was provided by US Department of Energy, Office of Fusion Energy Sciences. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Pseudo code for the DCE computational method

Algorithm 1 Calculate dislocation positions in a single DCE

```

1: Initialize:  $N \leftarrow 0$  // Number of crack dislocations
2: Initialize:  $N_{\min}, N_{\max}$  // Min/max number of crack dislocations
3: Initialize:  $\mathbf{x}, \mathbf{f}, \mathbf{b}$  // position, force, Burgers vector
4: Initialize:  $f_{\text{tol}}$  // force tolerance
5: Initialize: Step, Iter, MaxStep, MaxIter
6: Nucleate crack tip dislocations
7: while Step < MaxStep do
8:    $N_0 \leftarrow N$ 
9:   Iter  $\leftarrow 0$ 
10:  Nucleate a crack dislocation dipole
11:   $N \leftarrow N + 2$ 
12:  if  $N > N_{\max}$  then
13:    Increase  $|\mathbf{b}|$ 
14:    Delete all crack dislocations
15:     $N \leftarrow 0$ 
16:    Step  $\leftarrow$  Step + 1
17:    continue
18:  end if
19:  while Iter < MaxIter do
20:    Calculate  $\mathbf{x}$  using Velocity Verlet, FIRE, etc.
21:    if distance between dipole dislocations is too small then
22:      Delete these dislocations
23:       $N \leftarrow N - 2$ 
24:    end if
25:    Calculate  $\mathbf{f}$  using the stress field (applied + other dislocations)
26:    if  $|\mathbf{f}| < f_{\text{tol}}$  then
27:      break
28:    else
29:      Iter  $\leftarrow$  Iter + 1
30:      continue
31:    end if
32:  end while
33:  if Iter == MaxIter then
34:    Step  $\leftarrow$  Step + 1
35:    continue
36:  end if
37:  if  $N == N_0$  then
38:    if  $N > N_{\min}$  then
39:      break
40:    else
41:      Decrease  $|\mathbf{b}|$ 
42:      Delete all crack dislocations
43:       $N \leftarrow 0$ 
44:      Step  $\leftarrow$  Step + 1
45:      continue
46:    end if
47:  end if
48:  Step  $\leftarrow$  Step + 1
49: end while

```

Data availability

Data will be made available on request.

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